

ALL DIMENSIONS TO BE MEASURED OUT AND PEGGED ON SITE BEFORE COMMENCING WORK

ALL REINFORCED CONCRETE SLABS, FOUNDATIONS, COLUMN DETAILS, BEAMS, STAIRS AND RETAINING WALLS TO BE BUILT STRICTLY IN ACCORDANCE WITH ENGINEERS DETAIL AND UNDER HIS/HER SUPERVISION

BUILDING MUST BE IN ACCORDANCE WITH THE APPROVED BUILDING PLAN. NO CHANGES MAY BE MADE TO SUCH PLANS WITHOUT THE EXPRESSED CONSENT OF THE AUTHOR OF SUCH DOCUMENT

FLOORS, WALLS, GEYSER PIPES AND ROOFS TO BE INSULATED IN ACCORDANCE WITH SANS 10400

ARCHITECTURAL DRAWINGS TO BE READ IN CONJUNCTION WITH ENGINEERING DRAWINGS

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ACTS OF PARLIAMENT

All Contractors shall ensure that, before any work is put in hand, they comply with all the necessary Acts of Parliament of the Republic of South Africa.

ECO FLEX INSULATION TO HOT WATER PIPES TO SANS 10177-9 AND SANS 1077-10

ALL HOT WATER PIPES TO BE INSULATED TO ACHIEVE AN R-VALUE OF 1

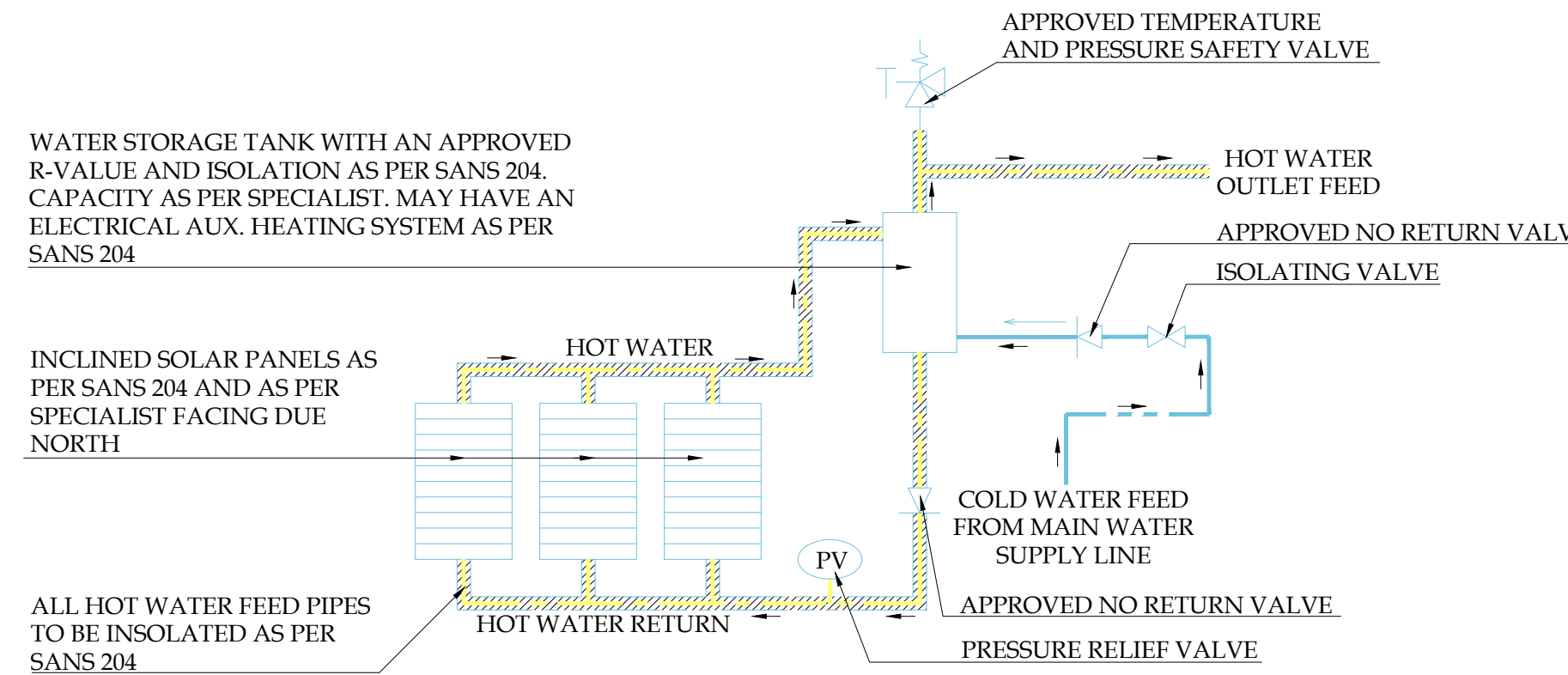
HOT WATER SYSTEM BE BE MAINTAINED IN ACCORDANCE WITH SANS 10252-1

SOLAR WATER HEATING SYSTEM TO SANS 1307 AND 10106. THERMAL PERFORMANCE TO BE DETERMINED IN ACCORDANCE WITH SANS 6211-1 AND 6211-2. INSTALLATION THEREOF WILL BE IN ACCORDANCE WITH SANS 10254

HOT WATER VESSELS AND TANKS TO HAVE A MIN R VALUE OF 2

PIPING TO BE INSULATED INCLUDES ALL FLOW AND RETURN PIPING, COLD WATER SUPPLY PIPING WITHIN 1M OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM AND PRESSURE RELIEF PIPING WITHIN 1M OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM. WHERE POSSIBLE, LENGTH OF PIPE RUNS SHOULD BE MINIMIZED

50% of HOT WATER TO COME FROM ALTERNATIVE SOURCE. GEYSER TO BE MAINTAINED AT 60° C



THERMOSIPHONIC HOT WATER SYSTEM DETAIL

SCALE N.T.S

HOT WATER SERVICES

A MINIMUM OF 50 % BY VOLUME OF THE ANNUAL AVERAGE HOT WATER HEATING REQUIREMENT SHALL BE PROVIDED BY MEANS OTHER THAN ELECTRICAL RESISTANCE HEATING, INCLUDING, BUT NOT LIMITED TO, SOLAR HEATING, HEAT PUMPS, HEAT RECOVERY FROM OTHER SYSTEMS OR PROCESSES ETC.

THE SOLAR WATER HEATING SYSTEMS SHALL COMPLY WITH SANS 1307 AND SANS 10106, BASED ON THE THERMAL PERFORMANCE DETERMINED IN ACCORDANCE WITH THE PROVISIONS OF SANS 6211-1 AND SANS 6211-2. THE INSTALLATION THEREOF SHALL COMPLY WITH SANS 10254.

HOT WATER USAGE SHOULD BE MINIMIZED AND THE SYSTEM MAINTAINED IN ACCORDANCE WITH THE REQUIREMENTS GIVEN IN SANS 10252-1.

ALL EXPOSED PIPES TO AND FROM THE HOT WATER CYLINDERS AND CENTRAL HEATING SYSTEMS SHALL BE INSULATED WITH PIPE INSULATION MATERIAL WITH AN R-VALUE AS FOLLOW:
A PIPE WITH AN INTERNAL DIAMETER SMALLER THAN 80mm = R-VALUE OF 1.00
A PIPE WITH AN INTERNAL DIAMETER LARGER THAN 80mm = R-VALUE OF 1.50

INSULATION SHALL BE PROTECTED AGAINST THE EFFECTS OF WEATHER AND SUNLIGHT, BE ABLE TO WITHSTAND THE TEMPERATURES WITHIN THE PIPING AND ACHIEVE THE MINIMUM TOTAL R-VALUE GIVEN ABOVE.

HOT WATER VESSELS AND TANKS SHALL BE INSULATED WITH A MATERIAL ACHIEVING A MINIMUM R-VALUE OF 2.0. NOTE TO ACHIEVE THIS VALUE, INSULATION IN ADDITION TO THE MANUFACTURERS' INSTALLED INSULATION MAY BE REQUIRED.

INSULATION ON VESSELS, TANKS AND PIPING CONTAINING COOLING WATER SHALL BE PROTECTED BY A VAPOR BARRIER ON THE OUTSIDE OF THE INSULATION.

THE PIPING INSULATION REQUIREMENTS DO NOT APPLY TO SPACE HEATING WATER PIPING LOCATED WITHIN THE SPACE BEING HEATED WHERE THE PIPING IS TO PROVIDE THE HEATING TO THAT SPACE OR ENCASED WITHIN A CONCRETE FLOOR SLAB OR IN MASONRY. THESE PIPES SHALL COMPLY WITH SANS 10252-1.

PIPING TO BE INSULATED INCLUDES ALL FLOW AND RETURN PIPING, COLD WATER SUPPLY PIPING WITHIN 1m OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM AND PRESSURE RELIEF PIPING WITHIN 1m OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM. WHERE POSSIBLE, LENGTHS OF PIPE RUNS SHOULD BE MINIMIZED.

SANS 10252-1 WATER LEGEND

—	Pipe carrying cold water
—	Pipe carrying hot water
—HR—	Pipe carrying hot water (return)
—M—	Pipe carrying mixed (temperature) water
—F—	Pipe carrying water for fire-fighting
—/—	Pipe crossing (not connecting)
—	Lagged pipe
—	Normal direction of flow
—+—	Pipe joint (butt welded)
—+—	Pipe joint (compression)
—+—	Pipe joint (flanged and bolted)
—+—	Pipe joint (screwed)
—+—	Pipe joint (soldered or solvent welded)
—+—	Pipe junction
50, 40	Pipe : nominal diameter (mm)
—	Pipe support (adjustable)
—	Pipe support (adjustable hanger)
—	Pipe support (flexible)
—	Pipe support (flexible hanger)
—	Pipe support (guide)
—	Pipe support (simple)
—	Pipe support (simple hanger)
●	Riser pipe (plan view)
○	Dropper pipe (plan view, pipe cross section)
—	Tap (external)
—	Tap (internal)
—	Mixer (thermostatically controlled)
—	Mixer (single manual controls, single lever)
—	Mixer (two manual controls)
—	Drinking fountain
—	Draining tap
—	Shower (fixed)
—	Shower (moveable)
—	Stop cock
—	Water meter
—	Fire brigade inlet/booster pump connection
—	Fire-hose reel

—	Irrigation sprinkler
—	Storage water heater (domestic type)
—	Expansion control valve
—	Expansion control valve (incorporating vacuum relief)
—	Expansion control vessels
—	Non-return valve
—	Orifice plate control device
—	Air release valve
—	Air vessel (air chamber)
—	Automatic shut-off valve
—	Balancing device (hot water control)
—	Double check valve/backflow preventer
—	Thermostatic controller
—	Vacuum relief valve
—	Vacuum relief valve combined with air release valve
—	Temperature and pressure safety valve (safety valve)
—	Pressure control valve
—	Pressure control valve (adjustable)
—	Pressure control valve combined with expansion control valve
—	Pressure control valve (adjustable)
—	Pressure relief valve
—	Pressure switch
—	Strainer
—	Float valve
—	Flow restrictor (controller)
—	Flow switch
—	Gauge (pressure:P ; Temperature:T ; Level:L)
—	Gauge with isolating valve
—	Isolating coupling for dissimilar metals
—	Isolating valve (flanged ends) (electrical control)
—	Isolating valve (flanged ends) (manual control)
—	Isolating valve (screwed ends) (electrical control)
—	Isolating valve (screwed ends) (manual control)
—	Water level control valve
—	Solar Heater
—	Heat Pump
—	Fire hydrant

1. GENERAL

- 1.1 THIS DRAWING IS NOT TO BE SCALED, USE FIGURED DIMENSIONS ONLY. ALL DIMENSIONS AND HEIGHTS TO BE CHECKED AND VERIFIED BEFORE ANY WORK COMMENCES ON SITE. ANY DISCREPANCIES SHALL BE REPORTED TO THE ARCHITECT IMMEDIATELY. ALL LEVELS, HEIGHTS OF PLINTHS, DEPTH OF EXCAVATIONS AND NUMBER OF STEPS TO BE FINALLY CHECKED BY CONTRACTOR ON SITE.
- 1.2 THE SITE TO BE TREATED IN ACCORDANCE WITH S.A.B.S. CODE OF PRACTICE NO. 0124-1977 WITH SHELLDRITE TERMITETRUPE SOIL POISONER.
- 1.3 TOP OF FOUNDATIONS TO BE A MINIMUM OF 300mm BELOW NATURAL GROUND LEVEL.
- 1.4 BACKFILL TO FOUNDATIONS.
- 1.5 TOP OF 75mm CONCRETE SURFACE BED TO BE MINIMUM OF 150mm ABOVE FINISHED GROUND LEVEL.
- 1.6 75mm THICK CONCRETE SURFACE BED TO BE ON 'GUNPLAS' USB GREEN DAMP PROOF MEMBRANE ON SAND BINDING ON WELL COMPACTED FILL.
- 1.7 GUNDLIF BRICKGRIP S.A.B.S. EMBOSSED D.P.C 375 MIC. UNDER ALL WALLS, WINDOW CILLS AND AT CHANGES IN FLOOR LEVELS.
- 1.8 STORMWATER SHALL BE REMOVED FROM DWELLING, YARD AND SITE.
- 1.9 ELECTRICAL INSTALLATION SHALL BE 'HEINEMANN' EARTH LEAKAGE SYSTEM IN DISTRIBUTION BOARD.
- 1.10 ALL BUILDING WORK TO BE CARRIED OUT IN ACCORDANCE WITH LOCAL AUTHORITIES BUILDING BY-LAWS AND REGULATIONS.

2. FOUNDATIONS

- 2.1 ALL FOUNDATIONS, RETAINING WALLS, STRUCTURAL CONCRETE WORK AND SUB SOIL STORMWATER DRAINAGE TO CIVIL ENG'S SPECIFICATIONS.
- 2.2 ALL SOIL COMPACTION TO CIVIL ENG'S SPECIFICATIONS.
- 2.3 ALL FOUNDATIONS TO BE REINFORCED AS REQUIRED BY ENGINEER.

3. FLOORS

- 3.1 MIN. 75mm CONCRETE SURFACE BED.
- 3.2 25mm CEMENT SCREED TO ALL FLOORS.
- 3.3 SKIRTING TO BE PROVIDED.

4. WALLS

- 4.1 220mm BRICK WALLS WITH GUNDLIF BRICKGRIP DPC.
- 4.2 BRICKFORCE EVERY 4 COURSES AND EVERY COURSE FOR 4 COURSES OVER OPENINGS.
- 4.3 ALL BRICKWORK MIN. 7 mPa CLASS II.
- 4.4 WINDOWS BUILT IN WITH GUNDLIF BRICKGRIP DPC.
- 4.5 ALL PAINTWORK TO COMPLY WITH MANUFACTURER'S SPECIFICATIONS WITH NECESSARY UNDERCOATS.
- 4.6 EXTERNAL-TO PLAN INTERNAL-PLASTER AND PAINT.

ALL LEVELS, DIMENSIONS, DEPTHS OF EXCAVATIONS, HEIGHTS OF PLINTHS, NUMBER OF STEPS AND SIZES OF FOUNDATIONS TO BE PLACED AT A HEIGHT OF EXISTING STRUCTURE

5. ROOF A

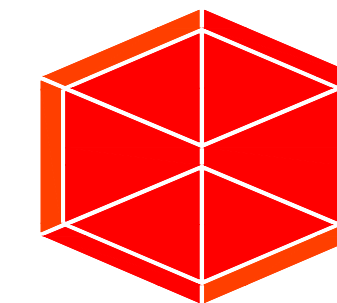
- 5.1 CONCRETE ROOF TILES @ 26° ON 38x38mm GRADED S.A. PINE TIMBER BATTENS @ 320mm C/C MAX ON 225x38mm PRE-FABRICATED, AS PER ENGINEER, S.A. PINE TRUSSES @ 760mm C/C MAX ON 114x38mm TIMBER WALL PLATES. TRUSSES TO BE ANCHORED WITH TWO STRANs OF 25mm GALVANIZED STEEL WIRE FIXED 300mm DEEP IN BRICK WALL. PROVIDE SABS APPROVED PLASTIC UNDERLAY AND ISALATION INSULATION.
- 5.2 MIN. 170mm CONCRETE SLAB TO ENGINEER'S DETAIL AND DESIGN.
- 5.3 MIN. 55mm VEMICULITE INSULATING SCREED TO 1:60 FALL.
- 5.4 WATERPROOFING BY SPECIALIST TO MANUFACTURERS SPEC.
- 5.5 1500 P.V.C RAINWATER DOWNPIPES FROM FULLBORE OUTLETS.
- 5.6 SKYLIGHTS TO MANUFACTURERS SPEC.
- 5.7 GUTTERS - MARLEY STREAMLINE SQUARE PROFILE RAINWATER GUTTERS + DOWNPIPES.
- 5.8 3mm SKM COAT RHINOLITE ON PLASTERBOARD CEILING FIXED TO 38x 38mm S.A. PINE BRANDING AT 450mm C/C FIXED TO TIE BEAMS.

6. WINDOWS AND DOORS

- 6.1 ANY PANE OF GLASS WHICH IS TO BE INSTALLED WITHOUT SUPPORT FRAME SHALL BE IN ACCORDANCE WITH SABS 0137
- 6.2 THICKNESS OF PANES IN RELATION TO THEIR AREA SHALL BE IN ACCORDANCE WITH SABS 0137
- 6.3 EXTERNAL SLIDING DOORS: STANDARD AS PER WINDOWS TO SCHEDULE.
- 6.4 INTERNAL DOORS: STANDARD HARDWOOD FRAMES TO SCHEDULE
- 6.5 WINDOWS: TO SCHEDULE
- 6.6 SLIDING DOORS: DOORS TO SCHEDULE.

7. DRAINAGE NOTES

- 7.1 1000 uPVC SEWER PIPE DRAIN WITH A MIN. FALL OF 1:60.
- 7.2 1000 OVP. AT HEAD OF DRAIN PIPE AS SHOWN.
- 7.3 RODDING EYES AT HEAD OF DRAIN, AT ALL CHANGES OF DIRECTION AND AT MAX. OF 25000 INTERVALS.
- 7.4 INSPECTION EYES AT ALL JUNCTIONS OF DRAIN, AND TO HAVE MARKED COVERS AT GROUND LEVEL.
- 7.5 DRAIN PIPE UNDER BUILDINGS TO BE PROTECTED AGAINST LOAD.
- 7.6 ALL WASTE PIPE TO HAVE 65mm RE-SEAL TRAPS, ALL WASTE PIPES TO BE ACCESSIBLE OVER ENTIRE LENGTH FOR CLEANING AND REPAIRS.
- 7.7 ALL WASTE PIPES UNDER FLOOR SLAB TO BE SLEEVED.
- 7.8 ALL SOIL FITTINGS WITH VERTICAL DISCHARGE GREATER THAN 1220 TO HAVE ANTISYPHON VENTPIPES.
- 7.9 RAINWATER DOWNPIPES TO DISCHARGE A MIN. OF 2440 FROM ANY
- 7.10 ALL DRAINAGE WORK TO BE CARRIED OUT IN ACCORDANCE WITH LOCAL AUTHORITIES DRAINAGE BY-LAWS AND REGULATIONS.



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STAND DESCRIPTION
NEW PROPOSED DWELLING ON
PTN 465 OF THE FARM
KROMDRAAI 560KQ
CLIENT:

WORKING DRAWINGS

TECHNOLOGISTS: W JUNG-KRUGER
ARCHITECT: W.R ELLIS

DATE: NOV 18 **DWG NO:**
DRAWN: WJK **SHEET:** 09 OF 10

GROUND STOREY WATER
SCALE 1:100

SPECIFICATION OF WATERPIPING MATERIAL:
PIPES > 75mm
CLASS 12 U PVC PIPES TO SABS 966 WITH FLEXIBLE JOINTS.
PIPES < 75mm
EITHER HDPE TYPE IV CLASS 12 TO SABS 533 WITH PLASSON JOINTS OR EQUAL (SUCH AS POLY PROPYLENE PIPE).

WATERPIPES SHALL HAVE A MINIMUM COVER OF 500MM
PROVISION SHOULD BE MADE FOR FUTURE CONNECTIONS IN ORDER TO MINIMIZE CUTTING INTO PIPES

GEYSER INSTALLATION TO
COMPLY WITH SABS 0254